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Fall 2025: CS 141 Exam #1

This is an **open-note** exam. You are permitted to use written or printed notes, but not a computer. The last page is a reference for the APIs that we've used in class; we suggest that you tear it off to make it easier to use during the exam. Students whose native language is not English may use a dictionary. *Nothing else is permitted.*

If a problem is unclear to you, include a brief explanation of how you interpret it.

Show your work!

You will not get partial credit if we cannot figure out how you got your answer.

Do not talk to other students or look at other students' exams.

Please sign below to indicate that you have upheld the highest standards of Knox College's Honor Code, and that you have not cheated in any way.

Sign here

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1. **(10) Terminology**, Next to each Java term given below, put the number of the phrase below corresponding most closely with its meaning. Each number will be used exactly once, but each row of the table may contain multiple numbers.

Variable declaration	
Constructor invocation	
A method that returns nothing	
When a Boolean expression in a condition always evaluates to true	
Converting one type to another	
A method that displays information given inside the parentheses on the computer monitor	
Defines a new reference data type	
Formal parameter	
A statement that adds 1	

```
1. for (int i = 0; 7 < 10; i++) {  
    ...  
}  
2. int x;  
3. System.out.println()  
4. new Turtle(new World());  
5. Scanner sc;  
6. public class World {  
    ...  
}
```

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7. `int myInt = 10;`
 `double myDouble = myInt;`
8. `CountNumber++;`
9. `int valueInt = (int) 25.5;`
10. `while(true){`
 `int x = 5;`
 `}`
11. `public static int max(int a, int b) {`
 `...`
 `}`
12. `x += 1;`
13. `public static void main(String[] args){`
 `...`
 `}`

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2. (10) **booleans**, Suppose that we have the following Java statements.

```
int aliens = 7;  
double spaceshipSpeed = 42.0;  
int planetsVisited = 3;  
String captainName = "Zorg";
```

Determine whether each of the following boolean expressions is (1) `true`, (2) `false`, or (3) a syntax error. If it is false, make a change that would make it true. If there is a syntax error, show how to fix it to make the statement true

- A. `aliens >= 5 && spaceshipSpeed < 50`
- B. `planetsVisited + aliens == 10`
- C. `captainName.substring(2, 3) == "R"`
- D. `spaceshipSpeed <> 42`
- E. `aliens % 2 = 1`
- F. `!(planetsVisited > aliens)`
- G. `spaceshipSpeed / 2 > 20 || aliens < 0`
- H. `captainName.charAt(2) == "r"`
- I. `planetsVisited * 2 <= aliens + 1`
- J. `captainName.length() == 4 && aliens < 10`

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3. **(10) Code errors**, Each of the 11 numbered lines in the following java program contain either one or zero syntax errors. For each of these 11 lines, either clearly fix the syntax error or write "None" next to the line if there are no errors.

1	<code>public class CodeTest</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int x == 10;</code>
4	<code> double y = 5,5;</code>
5	<code> String name = MyName;</code>
6	<code> if (x = 10) {</code>
7	<code> System.Out.println("X is 10");</code>
8	<code> } else</code>
9	<code> System.out.println('X is not 10');</code>
	<code> }</code>
10	<code> for (int i = 0; i < 5 i++) {</code>
11	<code> System.out.println("i = " + i);</code>
	<code> }</code>
	<code> }</code>
	<code>}</code>

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4. (20) **Code reading**, Show the output from each of the programs given below in the provided space.

a.

```
public class Prog1 {  
    public static void main(String[] args) {  
        int num = 4;  
        System.out.print(num);  
        num *= 2;  
        System.out.print(num);  
        num /= 4;  
        System.out.println(num);  
    }  
}
```

b. Assume the input types into the System.in is 141

```
import java.util.Scanner;  
public class Prog2 {  
    public static void main(String[] args) {  
        Scanner scan = new Scanner(System.in);  
        int score = scan.nextInt();  
        System.out.println(checkPass(score));  
    }  
  
    public static String checkPass(int s) {  
        if (s >= 100) {  
            return "Pass: " + s;  
        } else {  
            return "Fail: " + s;  
        }  
    }  
}
```

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c.

```
public class Prog3 {  
    public static void main(String[] args) {  
        System.out.println(checkNumber(7));  
        System.out.println(checkNumber(-2));  
    }  
  
    public static String checkNumber(int n) {  
        if (n > 0) return "Positive";  
        else if (n == 0) return "Zero";  
        else return "Negative";  
    }  
}
```

d.

```
public class Prog4 {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 3; i++) {  
            System.out.println(square(i));  
        }  
    }  
  
    public static int square(int x) {  
        int result = 0;  
        for (int j = 0; j < x; j++) {  
            result += x;  
        }  
        return result;  
    }  
}
```

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e.

```
public class Prog5 {  
    public static void main(String[] args) {  
        for (int a = 1; a <= 3; a++) {  
            for (int b = 1; b <= a; b++) {  
                System.out.println(a * b);  
            }  
        }  
    }  
}
```


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5. (10) Turtles, What does this Java Turtle code draw? Start at the little turtle image

```
Turtle t = new Turtle(world);
boolean leftTurn = true;
for (int i = 0; i < 5; i++)
{
    t.forward(100);
    if (leftTurn) {
        t.turn(60);
    } else {
        t.turn(-60);
    }

    t.forward(100);
    if (leftTurn) {
        t.turn(60);
    } else {
        t.turn(-60);
    }

    t.forward(100);
    if (leftTurn) {
        t.turn(60);
    } else {
        t.turn(-60);
    }
    // this switches the value of turn to its opposite
    // so from true to false and from false to true
    leftTurn = !leftTurn;
}
t.forward(100);
```



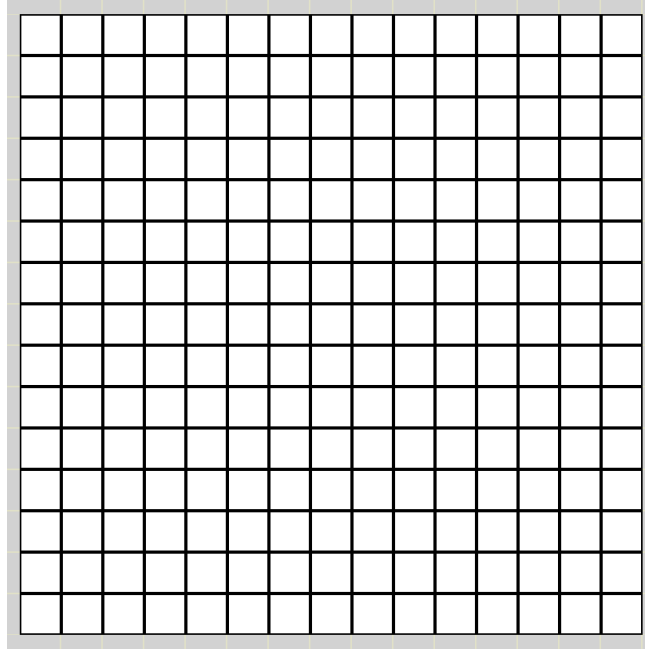
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6. **(10) Grades**, Write Java code that reads in an int value named `score` from a Scanner `scan` that has already been created for you below. Your program should print exactly one of the following messages based on the value of `score`:
If the score is 90 or above, print "Excellent!". If the score is 70 to 89, print "Good". If the score is 50 to 69, print "Pass". If the score is 1 to 49, print "Fail". If the score is 0 or negative, print "Invalid score". Your program should print **exactly one** of the 5 possible messages.

```
Scanner scan = new Scanner(System.in);  
System.out.println("Enter the score: " );
```

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7. **(20) Flags**, The grid will be at least 15x15. The function is already written for you; now start drawing and fill the grid boxes based on the following code!



```
public static void drawE(IDrawGrid g) {  
    // size here is the number of rows, but since  
    // the grid is always square, it is also the number of columns  
    int size = g.getNumRows();  
    for (int i=0; i < size; i++) {  
        g.setColor(i, i, Color.GRAY);  
        if (i > 0) {  
            g.setColor(i, i - 1, Color.GRAY);  
        }  
        if (i < size-1) {  
            g.setColor(i, i + 1, Color.GRAY);  
        }  
  
        g.setColor(i, size-1-i, Color.GRAY);  
        if (i > 0) {  
            g.setColor(i, size-i, Color.GRAY);  
        }  
        if (i < size-1) {  
            g.setColor(i, size-i-2, Color.GRAY);  
        }  
    }  
}
```

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8. (10) **Strings**, Complete the following code that takes a String named `str` as an input and prints the new String `result` with all the vowels ('a', 'e', 'i', 'o', 'u') removed. For example:

User Input:	Expected Output:
"abcdefghij"	"bcdfghj"
"minecraft"	"mncrft"
"spiderman homecoming"	"spdrmn hmcmmng"

```
public class RemoveVowels {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a string: ");  
        String str = sc.nextLine();  
        String result = "";  
        //Type your code here
```

```
        System.out.println(result);  
    }  
}
```

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Exam 1 Reference Sheet

Useful methods of `String`:

- `char charAt(int index)` returns the char at the specified index
- `boolean equals(Object s)` returns whether this `String` and `s` have the same value
- `int indexOf(String s)` returns index within this `String` where `s` first occurs (or -1 if `s` does not occur)
- `int length()` returns the number of characters in the `String`
- `String substring(int index)` returns `String` that has the contents of this `String` starting at given index
- `String substring(int s, int t)` returns `String` that has the contents of this `String` starting at index `s` and going to (but not including) index `t`
- `String toLowerCase()` returns result of converting all characters to lower case
- `String toUpperCase()` returns result of converting all characters to upper case
- `+` operator that takes two strings and gives the results of concatenating them (joining them)

Useful methods of `Turtle`:

- `Turtle(World w)` constructor that creates a `Turtle` drawing in the given `World`
- `void backward(int pixels)` causes `Turtle` to move backward a given number of pixels
- `void forward(int pixels)` causes `Turtle` to move forward a given number of pixels
- `double getHeading()` returns the direction the `Turtle` is currently facing (in degrees)
- `int getXPos()` returns `Turtle`'s current x coordinate
- `int getYPos()` returns `Turtle`'s current y coordinate
- `boolean isPenDown()` returns whether the pen is currently down
- `void penUp()` causes `Turtle` to lift its pen so it is no longer drawing
- `void penDown()` causes `Turtle` to lower its pen so it will draw while moving
- `void turn(int degrees)` causes the `Turtle` to turn clockwise by the given number of degrees

Useful methods of `IDrawGrid`:

- `int getNumCols()` returns the number of columns
- `int getNumRows()` returns the number of rows
- `void setColor(int row, int col, Color color)` sets the pixel at the given position to the given `Color`